Using our newfound appreciation of the subtleties of time, we realize that from Slim's perspective he is stationary while Jim is moving, and hence Slim sees Jim's clock as running slow. As a result, Slim realizes that Jim's indirect measurement of the car's length will yield a shorter result than he measured in the showroom, since in Jim's calculation (length equals speed multiplied by elapsed time) Jim measures the elapsed time on a watch that is running slow. If it runs slow, the elapsed time he finds will be less and the result of his calculation will be a shorter length.

Thus Jim will perceive the length of Slim's car, when it is in motion, to be less than its length when measured at rest. This is an example of the general phenomenon that observers perceive a moving object as being shortened along the direction of its motion. For instance, the equations of special relativity show that if an object is moving at about 98 percent of light speed, then a stationary observer will view it as being 80 percent shorter than if it were at rest. This phenomenon is illustrated in Figure 2.4.4

Motion through Spacetime

The constancy of the speed of light has resulted in a replacement of the traditional view of space and time as rigid and objective structures with a new conception in which they depend intimately on the relative motion between observer and observed. We could end our discussion here, having realized that moving objects evolve in slow motion and are foreshortened. Special relativity, though, provides a more deeply unified perspective to encompass these phenomena.

To understand this perspective, let's imagine a rather impractical automobile that rapidly attains its cruising speed of 100 miles per hour and sticks to this speed, no more, no less, until it is shut off and rolls to a halt.

Let's also imagine that, due to his growing reputation as a skilled driver, Slim is asked to test-drive the vehicle on a long, straight, and wide track in the middle of a flat stretch of desert. As the distance between the start and finish lines is 10 miles, the car should cover this distance in one-tenth of an hour, or six minutes. Jim, who moonlights as an automobile engineer, inspects the data recorded from dozens of test-drives and is disturbed to see that although most were timed to be six minutes, the last few are a good deal longer: 6.5, 7, and even 7.5